

Solving the following system:

E-Vex

$$\begin{cases} \ln x + \ln y = 0 \\ x + y = \frac{10}{3} \end{cases}$$

$$\ln x + \ln y \Rightarrow \ln(x \cdot y) = 0$$

$$e^{\ln(x \cdot y)} = e^0 \quad \text{then} \quad \begin{cases} x \cdot y = 1 \\ x + y = \frac{10}{3} \end{cases}$$

$$\boxed{y = \frac{10}{3} - x}$$

$$x\left(\frac{10}{3} - x\right) = 1$$

$$\frac{x \cdot 10}{3} - x^2 - 1 = 0 \xrightarrow{*-1} x^2 - \frac{x \cdot 10}{3} + 1 = 0$$

$$\text{then} \Rightarrow 3\left(x^2 - \frac{x \cdot 10}{3} + 1\right) = 3(0)$$

$$3x^2 - 10x + 3 = 0$$

$$x = 3, x = \frac{1}{3} \Rightarrow 0 < \frac{1}{3} < 3 \quad \underline{\text{acc}}$$

$$\text{When } x = 3: \begin{cases} 3y = 1 \\ y = \frac{1}{3} \end{cases}$$

$$\text{When } x = \frac{1}{3}: \begin{cases} \frac{1}{3}y = 1 \\ y = 3 \end{cases}$$

$$S: \left(3, \frac{1}{3}\right) \left(\frac{1}{3}, 3\right)$$

~~E-Vex~~